

Title: Cowrie Honeypot – Network Attack Detection & Reporting

Name Iwuji Chikaodi Edith

Date 9/01/2026

Executive Summary

This report documents a network attack detection exercise conducted using a Cowrie SSH honeypot in a controlled virtual lab environment. The objective was to simulate attacker behavior, monitor unauthorized access attempts, and analyze network traffic generated during the interaction. Using tools such as Nmap and Wireshark, open services were identified and SSH-based attack activity was captured and analyzed. The results demonstrate the effectiveness of honeypots in detecting suspicious activity without exposing real production systems, highlighting their value in proactive cybersecurity monitoring and incident response.

Objective

The objective of this exercise was to deploy a Cowrie SSH honeypot on an Ubuntu server, perform network reconnaissance from a Kali Linux machine, capture attacker behavior using Wireshark, and document all findings in a structured incident report. The goal was to practice hands-on network security monitoring, honeypot configuration, and incident documentation.

Lab Environment

- Attacker Machine: Kali Linux
- Target Machine: Ubuntu Server running Cowrie Honeypot
- Virtualization: VMware

- Network: Local VM Network (192.168.37.0/24)
- Tools: Cowrie Honeygot, Nmap, Wireshark

Honeygot Deployment:

The Cowrie SSH honeygot was deployed on an Ubuntu Server virtual machine, The honeygot was started manually using the twisted framework with the command “twisted -n cowrie” , allowing real-time monitoring of attacker interactions

Network Scanning & Detection:

The attacker machine {Kali Linux} performed reconnaissance on the target system using Nmap, The scan identified active SSH services running on ports 22 and 2222, the commands used are:

- nmap 192.168.37.135
- Nmap -sV 192.168.37.135
- Nmap -A 192.168.37.135

Port 22/tcp was open and running a legitimate OpenSSH service

Port 2222/tcp was open and emulated an SSH service by the cowrie honeygot.

Service version detection revealed SSH service designed to appear legitimate.

OS fingerprinting identified the target as a Linux-based system.

Attack Simulation & Interaction

An SSH connection was initiated from the Kali Linux machine to the target system on port 2222, which is monitored by the cowrie honeygot. The login attempt successfully connected to a simulated shell environment. Several commands {root,admin,test} were entered to simulate attacker post-exploitation behaviour. These commands were logged by the honeygot but did not execute on the real system, confirming successful attack containment

Wireshark Capturing

Network traffic was captured using Wireshark during the SSH attack stimulation. The capture shows TCP communication between the attacker [Kali Linux] and the honeypot {Cowrie} over port 2222, including SSH handshake and authentication attempts.

Recommendations

- Deploy honeypots in production networks to detect early attacks
- Regular monitor SSH attempts and logs
- Conduct periodic vulnerability scanning to complement honeypot monitoring.
- Use captured attack data to update security policies and incident response playbooks

Conclusion

This lab exercise demonstrated practical skills in deploying and monitoring a honeypot, analyzing network attacks using Nmap and Wireshark, and documenting a cybersecurity incident professionally. The project reinforces key SOC, defensive security, and documentation skills for real-world cybersecurity operations